Ethanol Blending in Gasoline – Netherlands

September 2025

U.S. GRAINS & BIOPRODUCTS COUNCIL

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Ethanol Blending in Europe

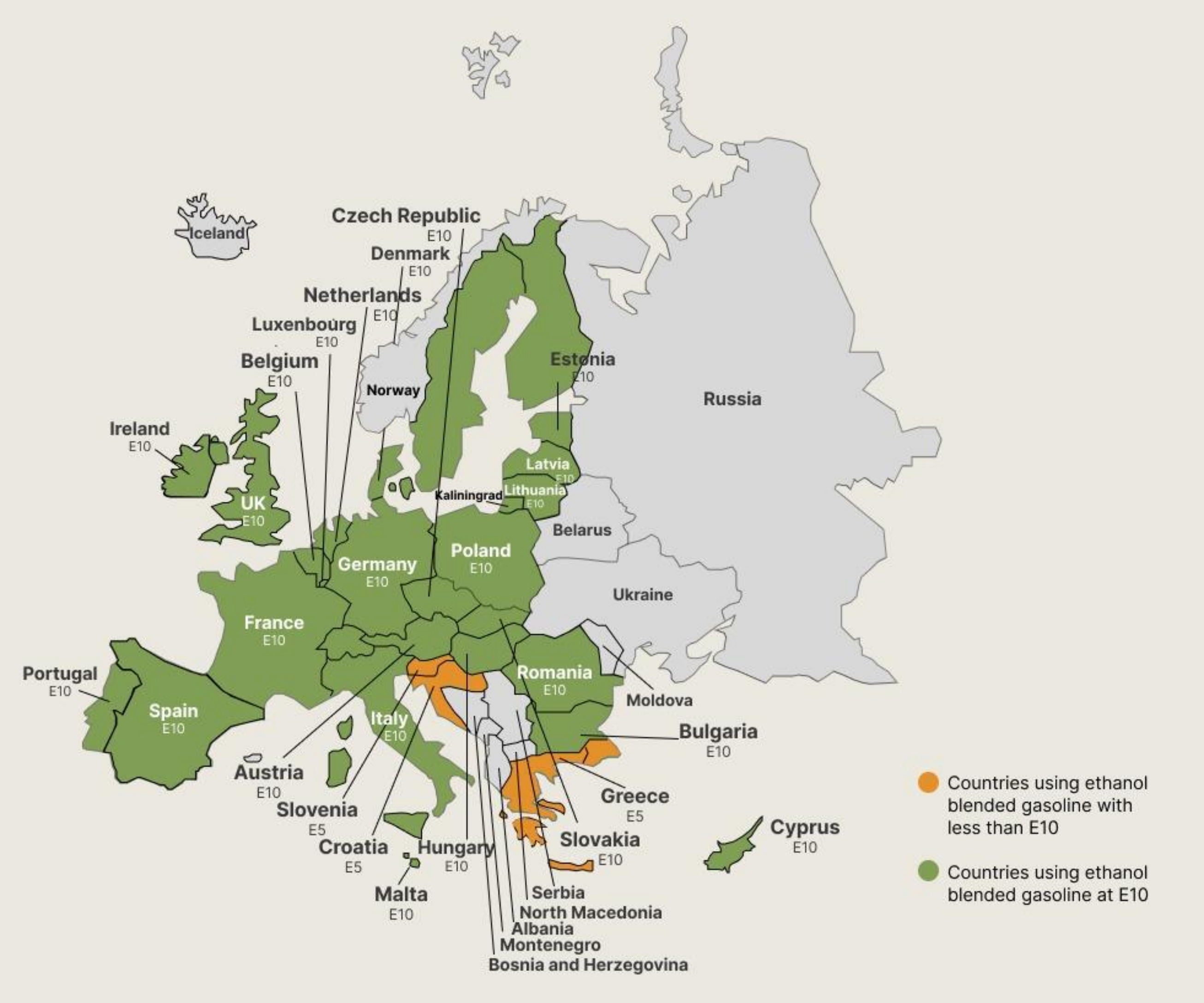
There are important fuel quality and environmental impact of vehicle emission challanges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.

- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.

- Ethanol contributes to transport decarbonization and air quality improvement.

- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

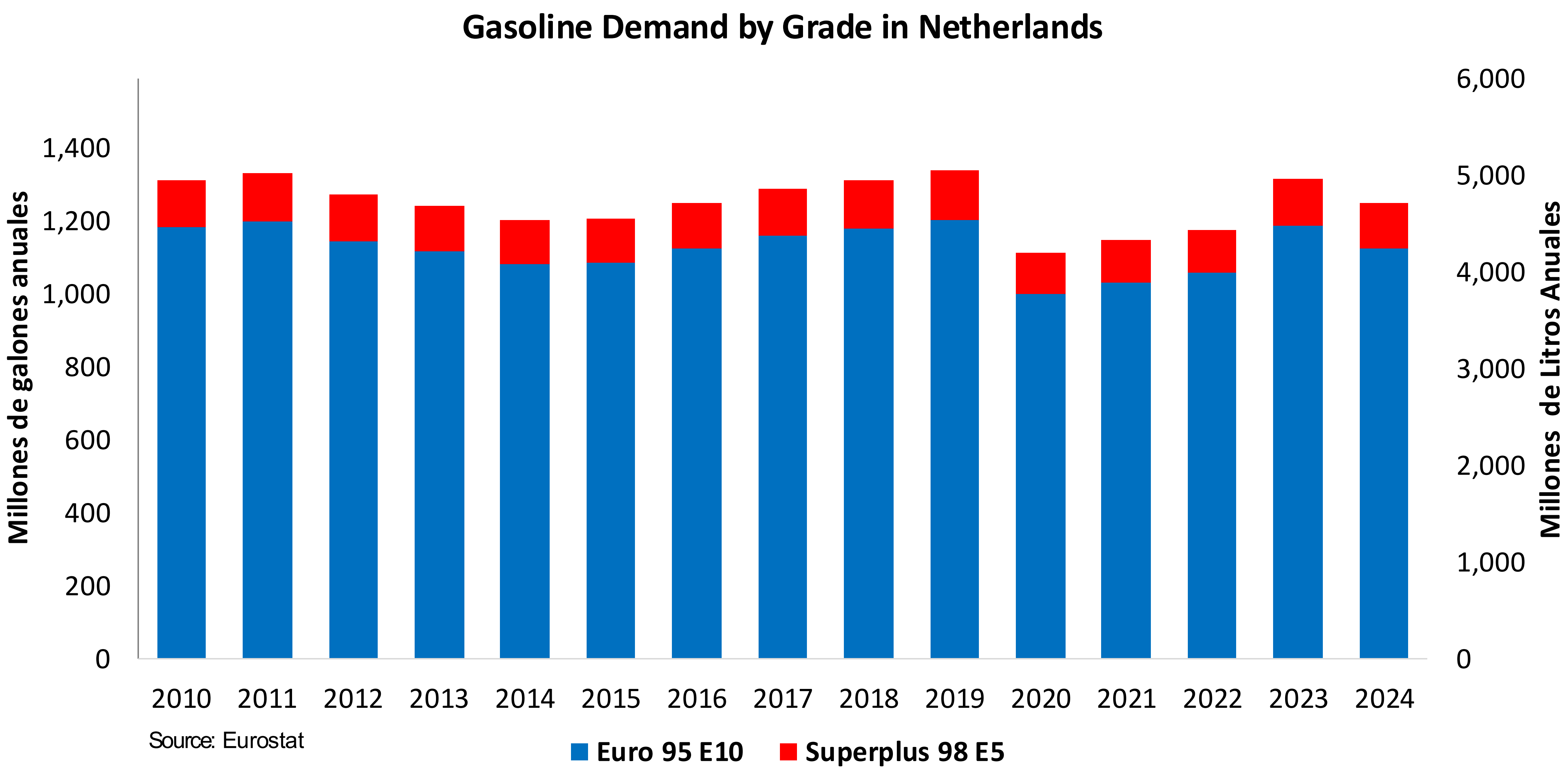
Ethanol Gasoline Blending in Netherlands

In 2024, gasoline consumption in Netherlands was 4,723 million liters (1,248 million gallons) and growth rate was -1.3%. Gasoline production and net imports were 6,295 and -14,373 million liters (1,663 and -3,797 million gallons) correspondingly. Netherlands complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

In 2024, ethanol consumption in Netherlands was 507 million liters (134 million gallons) representing 10.7% of gasoline demand. Ethanol production and Net Imports were 1,080 and -574 million liters (134 and -152 million gallons) correspondingly. Ethanol sources are Sugar Cane 5% and Cereals 95%.

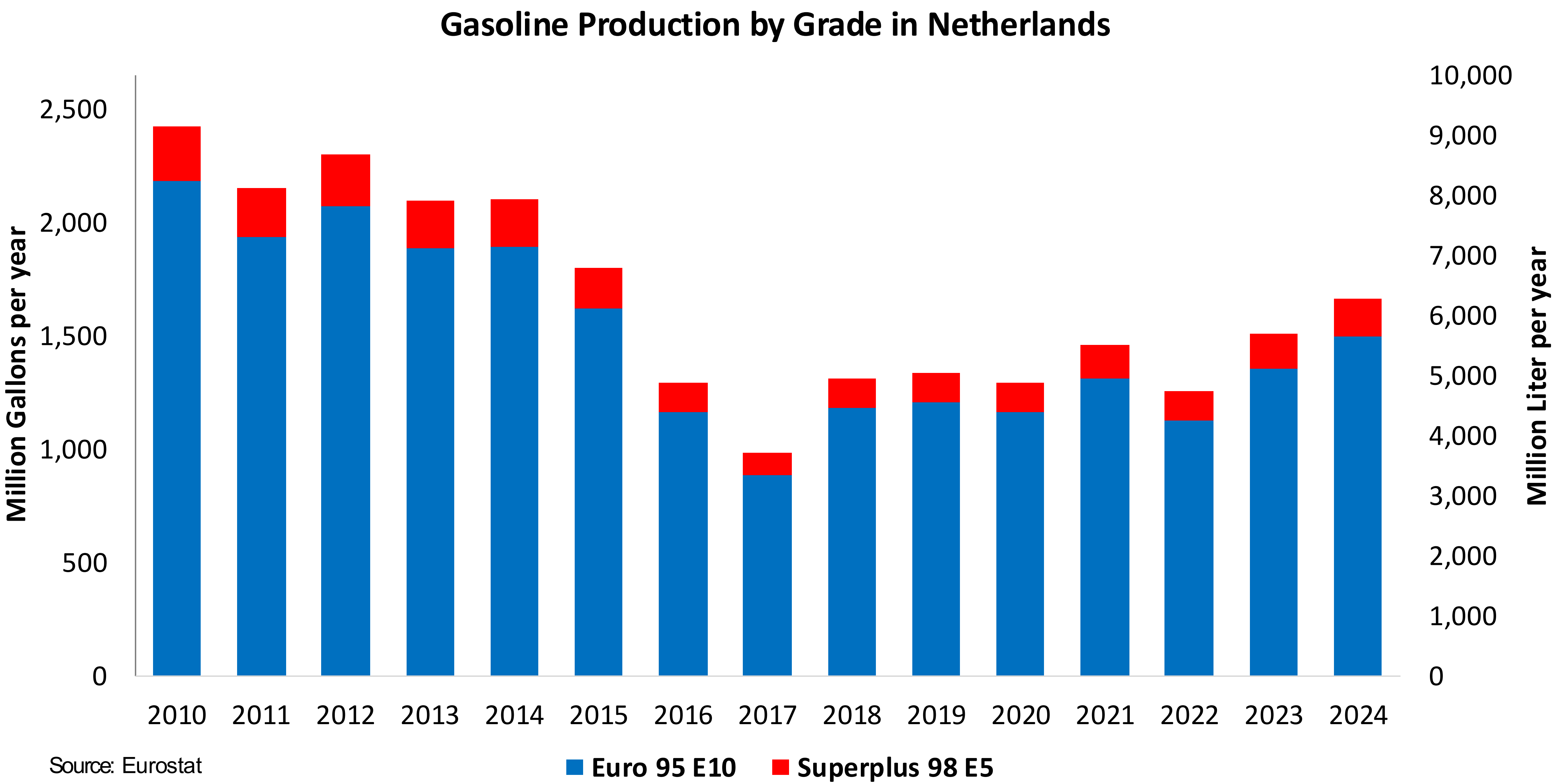
Source: Eurostat, European Commision

Gasoline Demand in Netherlands



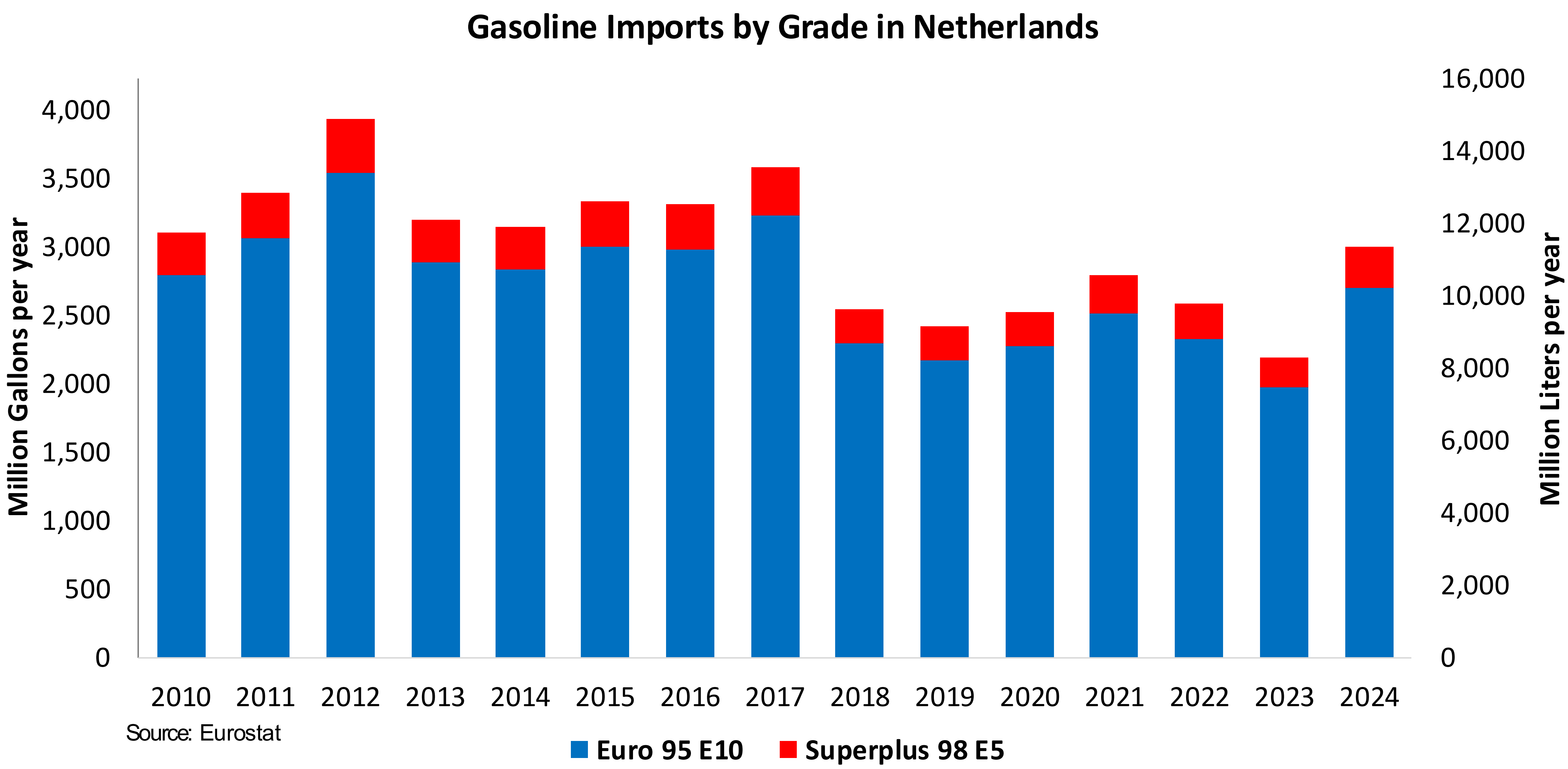
Source: Eurostat

Gasoline Production in Netherlands



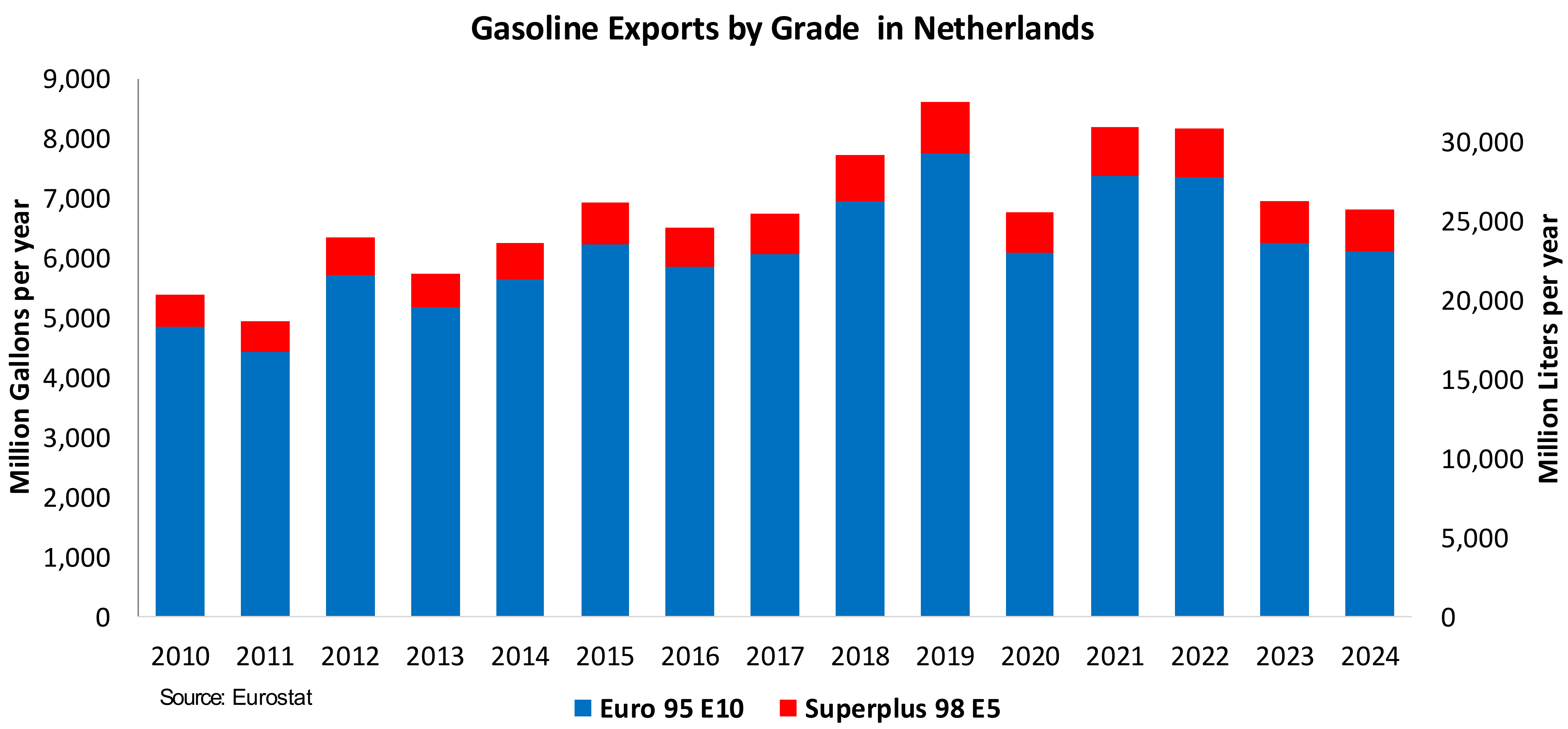
Source: Eurostat

Gasoline Imports in Netherlands

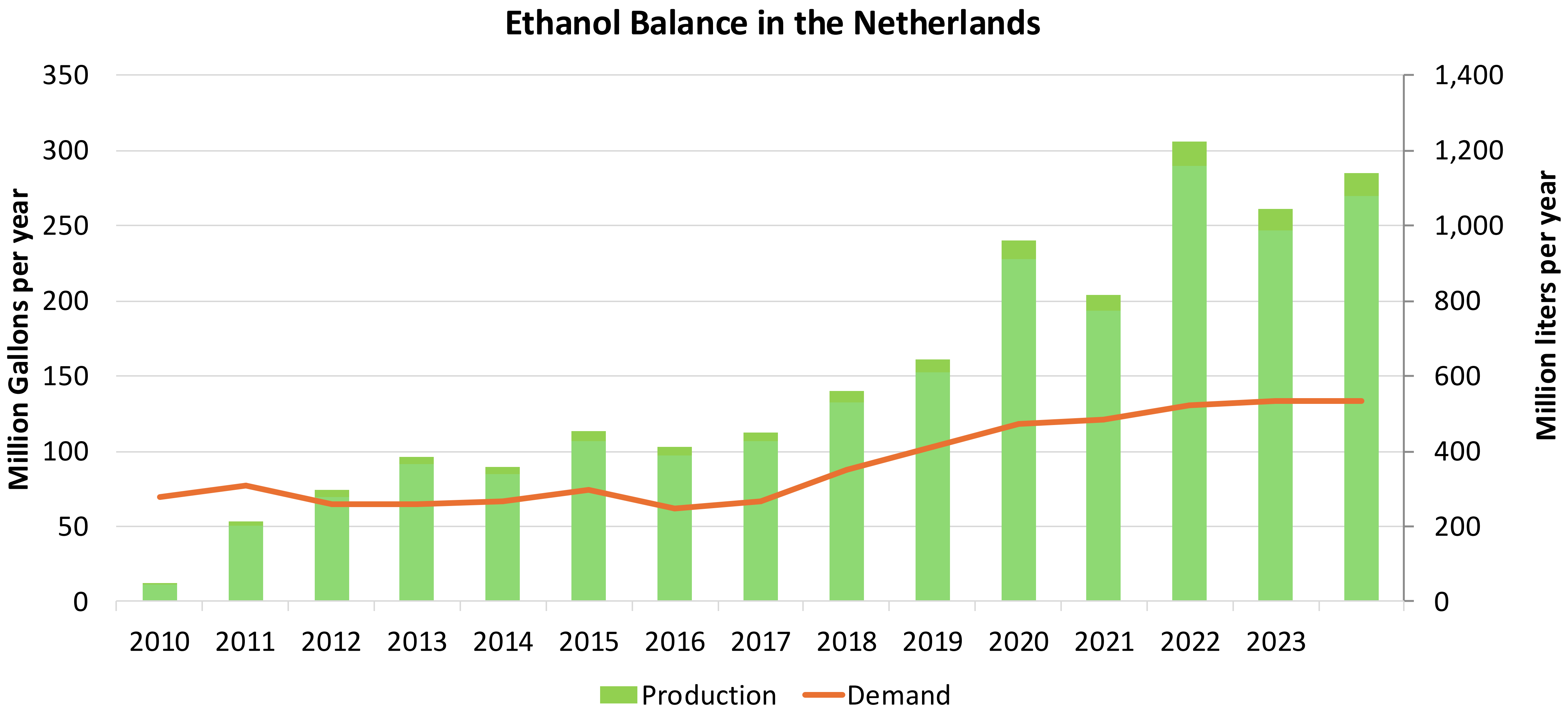


Source: Eurostat

Gasoline Exports in Netherlands



Ethanol Balance in Netherlands



Source: Eurostat, Statistics Netherlands

Gasoline Quality in Netherlands

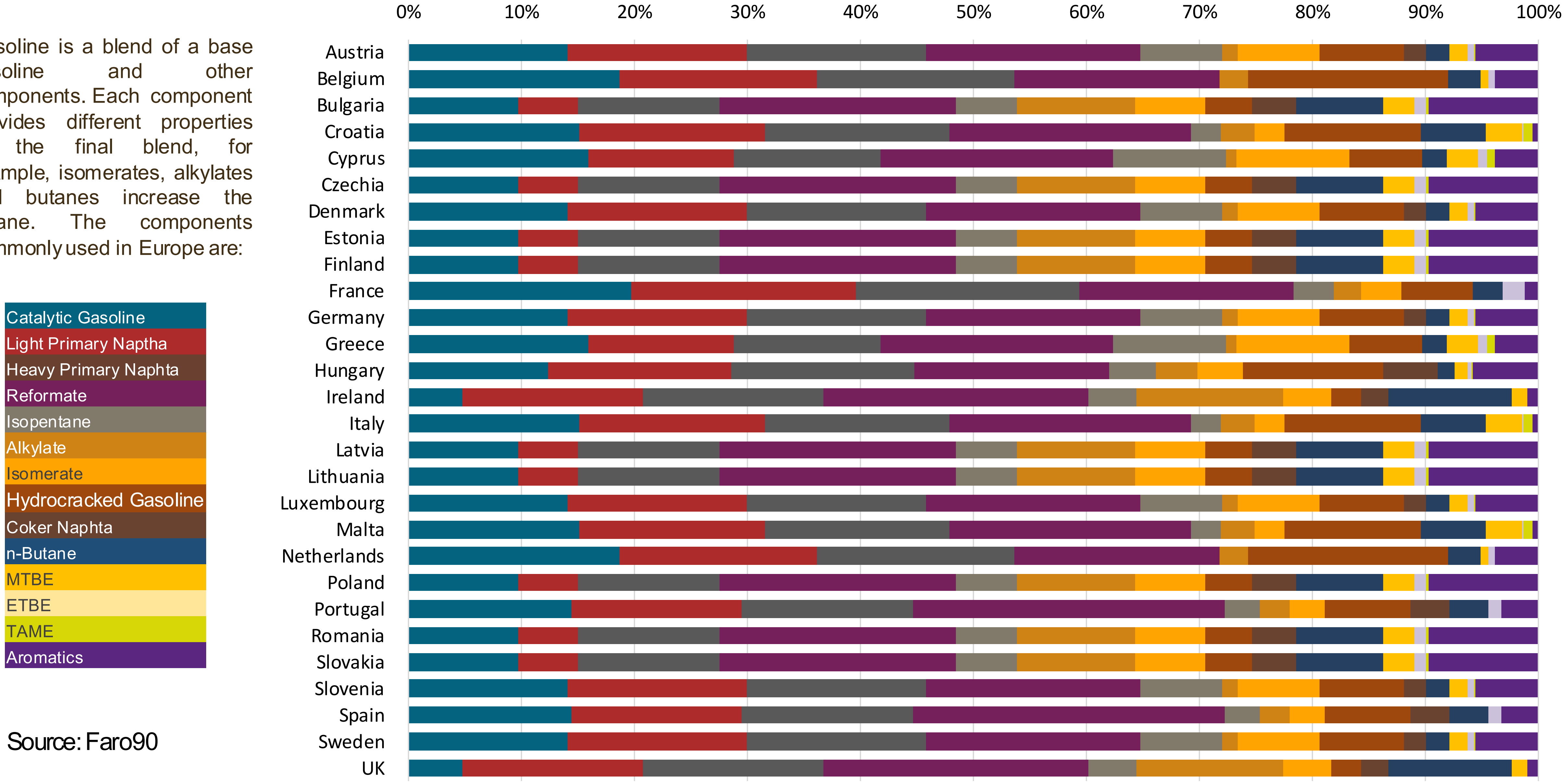
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	28.4							
Advanced Biofuels (%cal)	3.6							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



Source: Faro90

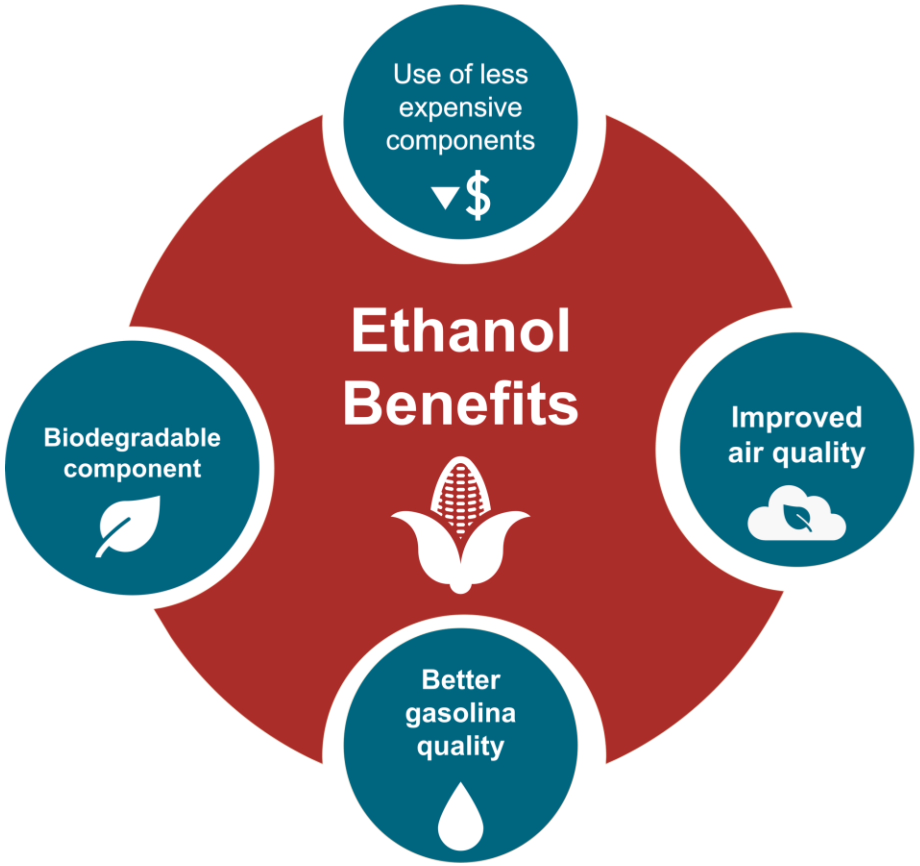
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

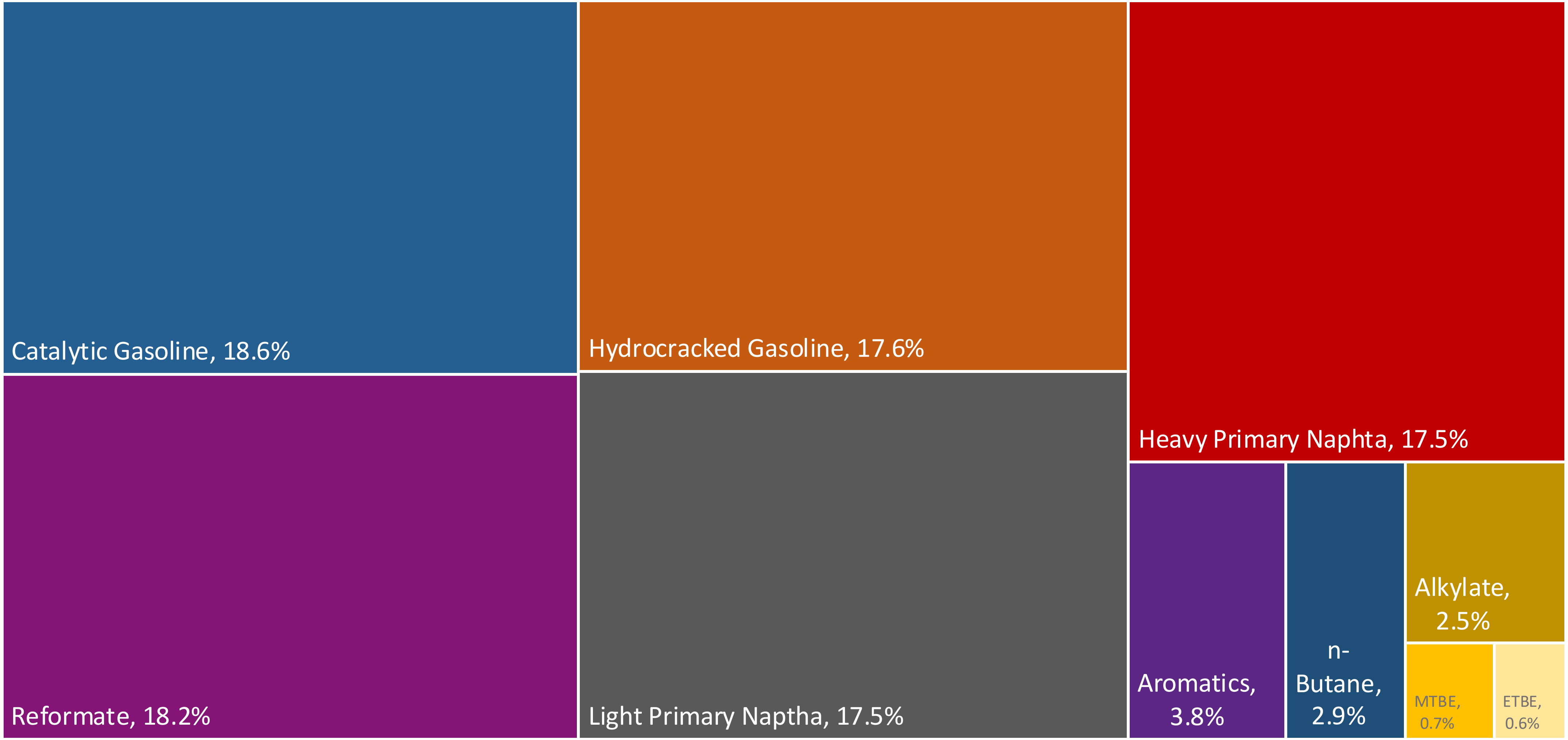
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

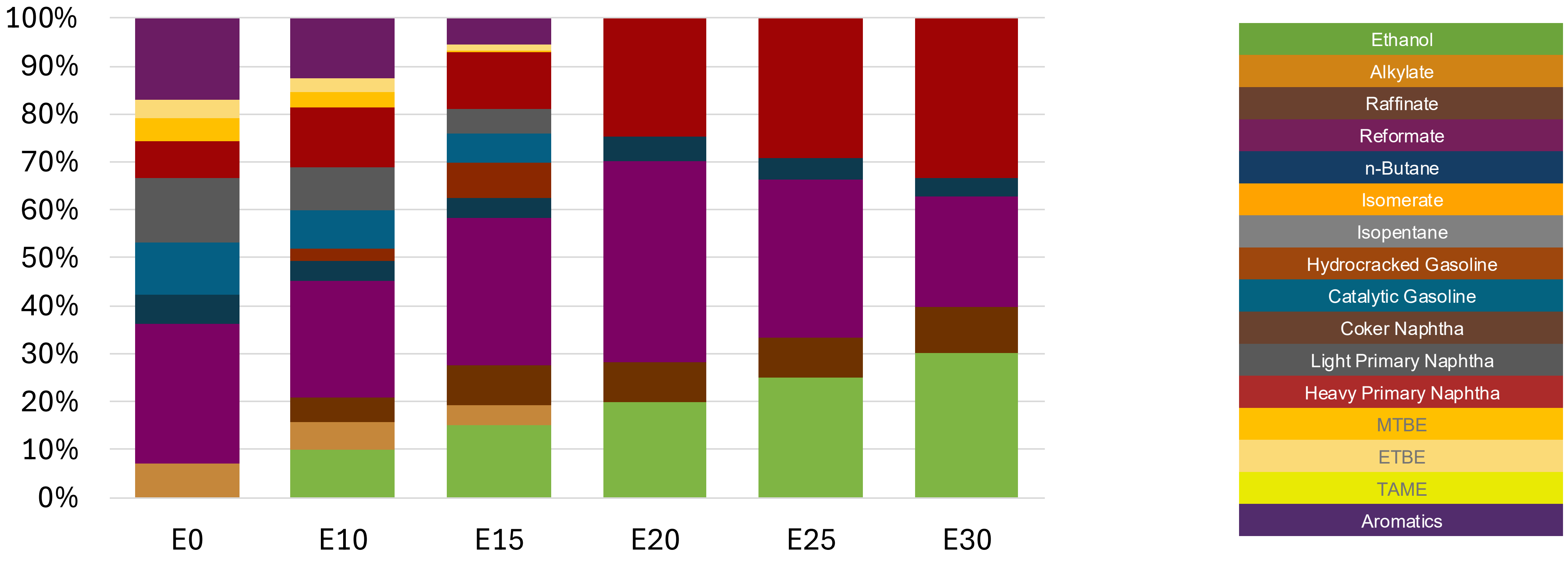


Netherlands Available Blending Components



Source: Faro90

Netherlands – Gasoline 95 – Constant Octane

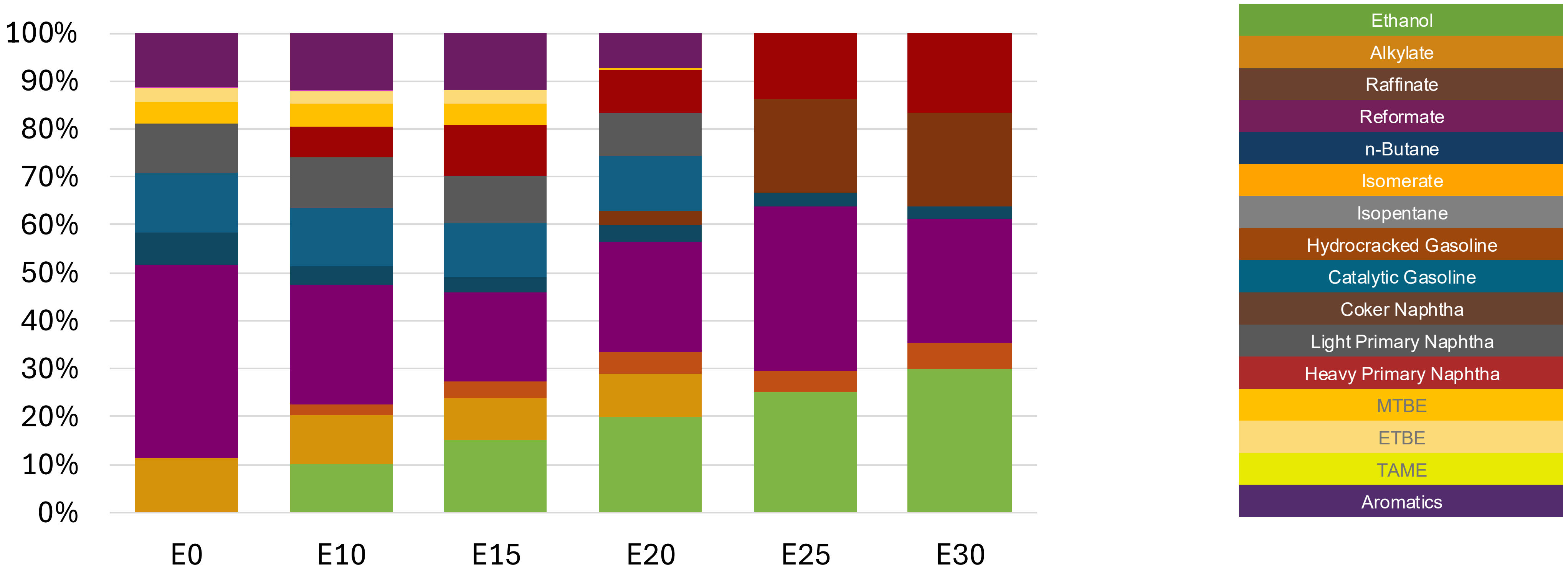


Octane (RON)	95.0	95.0	95.0	95.1	95.3	95.3
Price (US$/gal)	$ 2.281	$ 2.147	$ 2.072	$ 1.951	$ 1.875	$ 1.780

Average CIF 2024 Prices

Source: Faro90

Netherlands - Gasoline 98 - Constant Octane

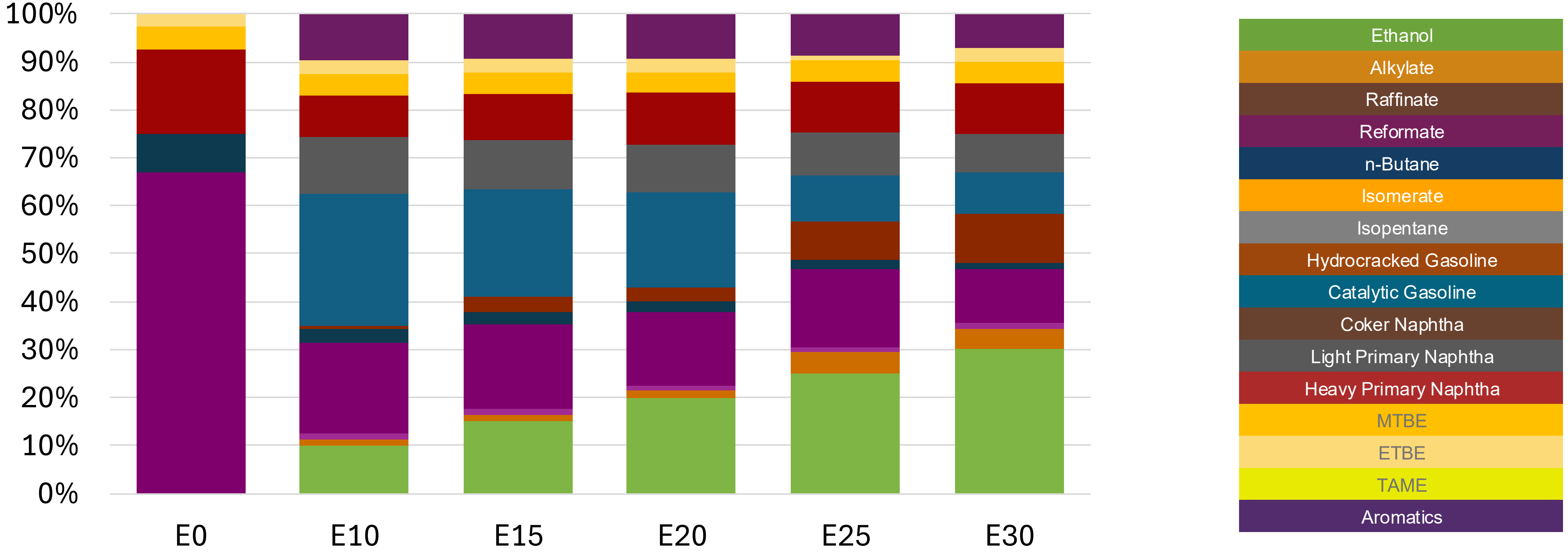


Octane (RON)	98.0	98.0	98.0	98.0	98.1	98.5
Price (US$/gal)	$ 2.409	$ 2.290	$ 2.216	$ 2.164	$ 2.057	$ 1.994

Average CIF 2024 Prices

Source: Faro90

Netherlands – Gasoline 95 – Increased Octane

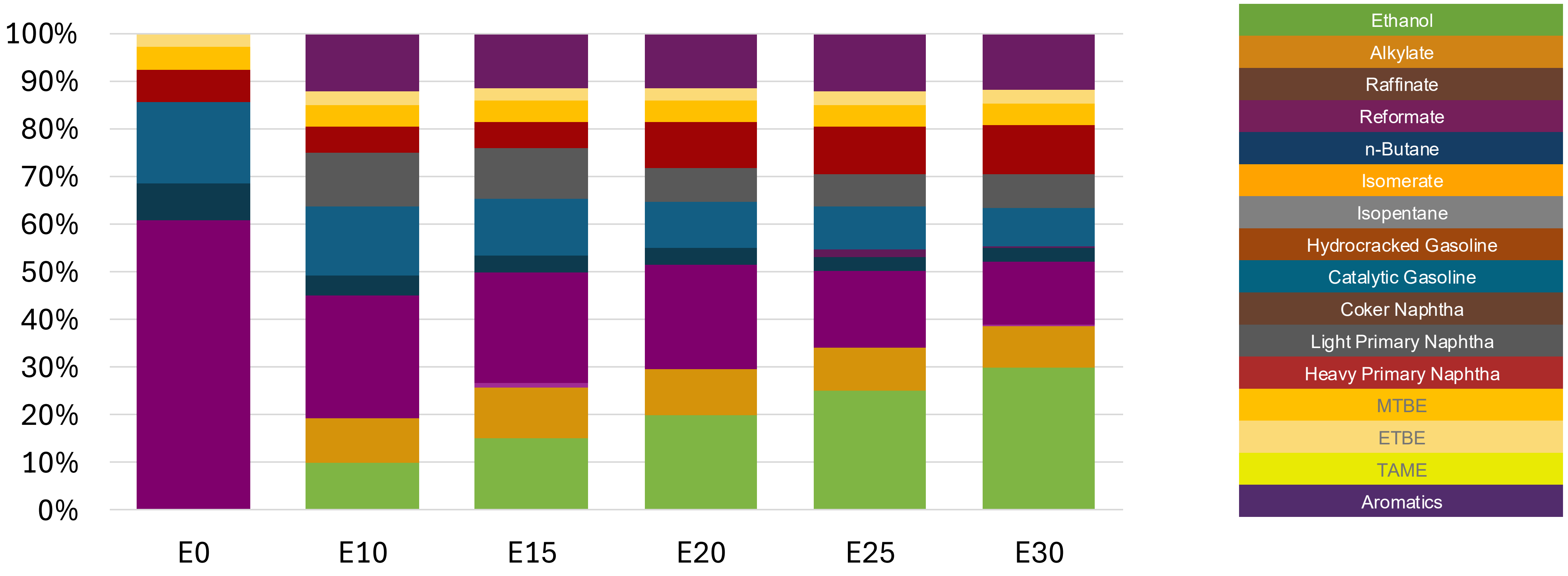


Octane (RON)	95.0	96.5	97.9	99.4	100.5	102.1
Price (US$/gal)	$ 2.193	$ 2.193	$ 2.193	$ 2.193	$ 2.193	$ 2.193

Average CIF 2024 Prices

Source: Faro90

Netherlands – Gasoline 98 – Increased Octane



Octane (RON)	98.0	98.8	100.1	101.6	102.9	104.2
Price (US$/gal)	$ 2.329	$ 2.329	$ 2.329	$ 2.329	$ 2.329	$ 2.329

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

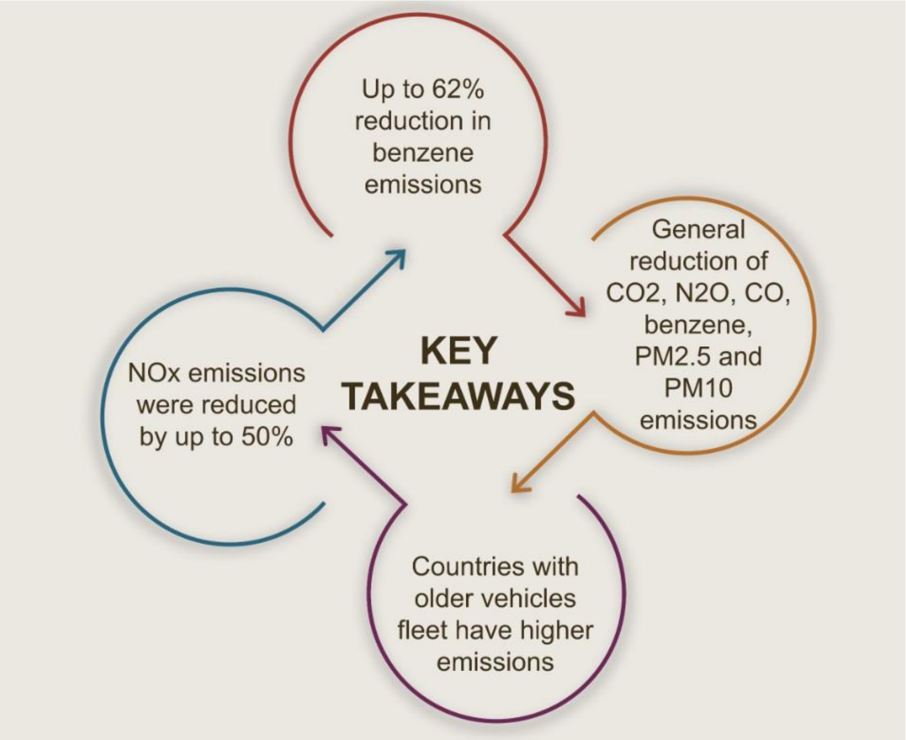
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

Gasoline Vehicle Fleet - Netherlands

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	6,216
		Single FI	3-Ways		>8 years	12,798
		Multi FI	3-Ways		>8 years	793,203
			Euro III		>8 years	96,743
			Euro IV		>8 years	700,482
			Euro V		>8 years	484,680
			Euro 6		>8 years	235,874
					4-8 years	134,841
					<4 years	200,369
			Hybrid		All	161,840
		Electric			All	19,143
	Medium	Carburator	3-Ways	PCV Tank	>8 years	615
		Single FI	3-Ways		>8 years	180
		Multi FI	3-Ways		>8 years	84,622
			Euro III		>8 years	26,441
			Euro IV		>8 years	236,181
			Euro V		>8 years	146,819
			Euro 6		>8 years	78,625
					4-8 years	67,627
					<4 years	158,736
			Hybrid		All	93,586
		Electric			All	11,070
Sml Engines	Light	4-Cycle	None	PCV	>8 years	103,199
			Euro III		>8 years	106,680
			Euro IV		>8 years	603,459
			Euro V		>8 years	358,694
			Euro 6		>8 years	858,679
					4-8 years	653,077
					<4 years	1,634,834
			Hybrid		All	42,874
		Electric			All	5,071
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	74,584
		Fuel-Injection	3-Ways		>8 years	842,906
			Euro III		>8 years	472,733
			Euro IV		>8 years	544,933
			Euro V		>8 years	152,548
			Euro 6		>8 years	210,599
					4-8 years	99,290
					<4 years	12,899
			Hybrid		All	14,445
		Electric			All	1,709

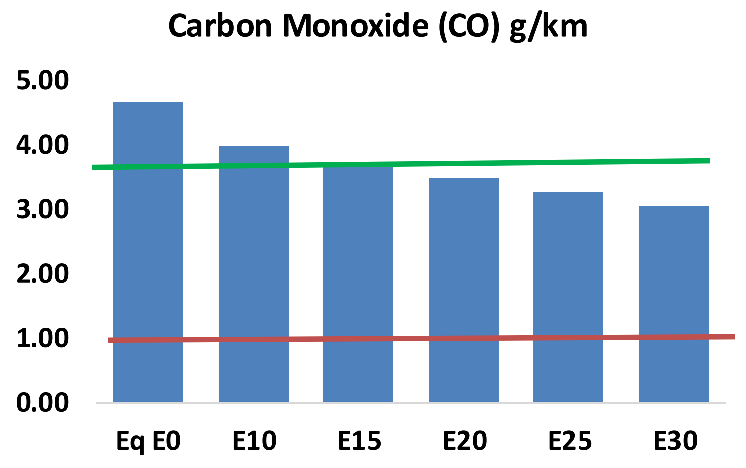
0 200,000 400,000 600,000 800,000 1,000,000 1,200,000 1,400,000 1,600,000 1,800,000

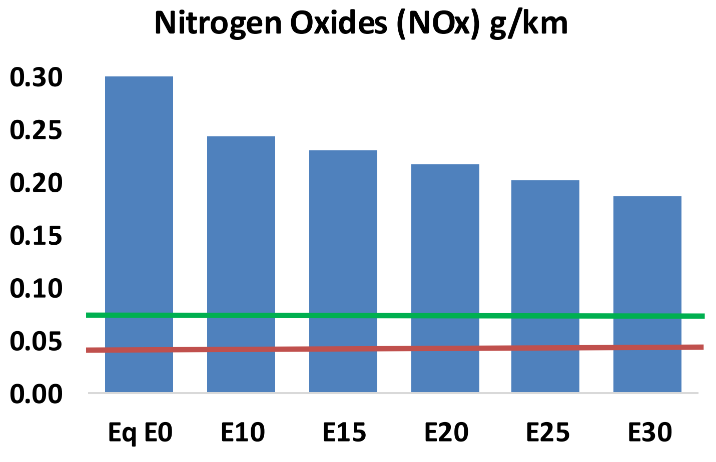
Vehicle Fleet: 10,593,903
Gasoline Vehicle Fleet: 7,721,374

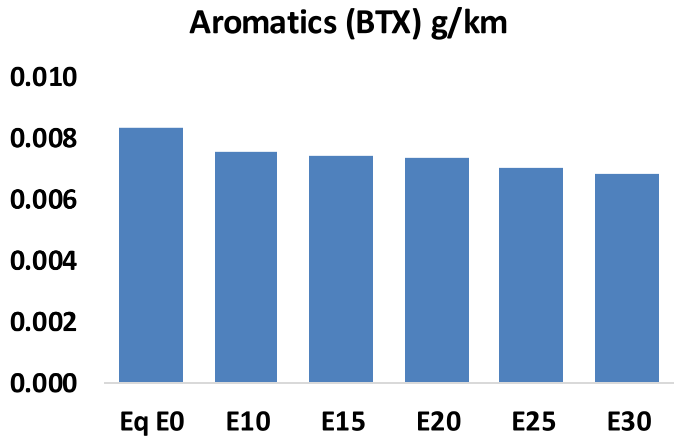
Source: Eurostat, ACEA

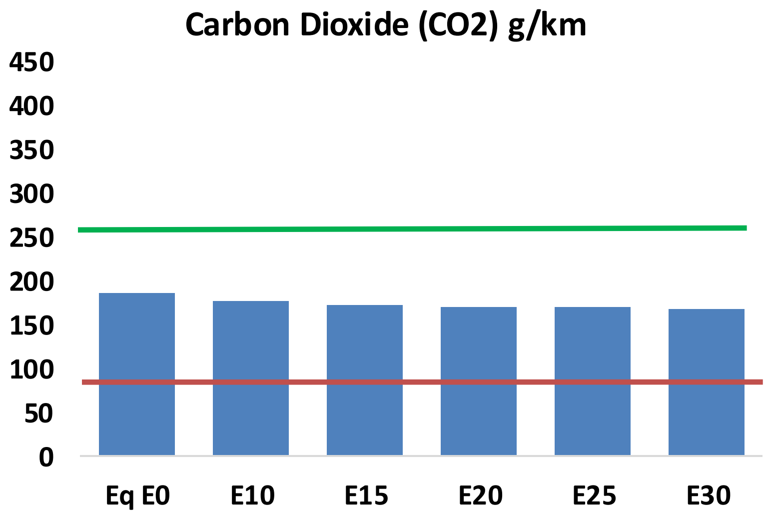
Netherlands – Vehicular Emissions

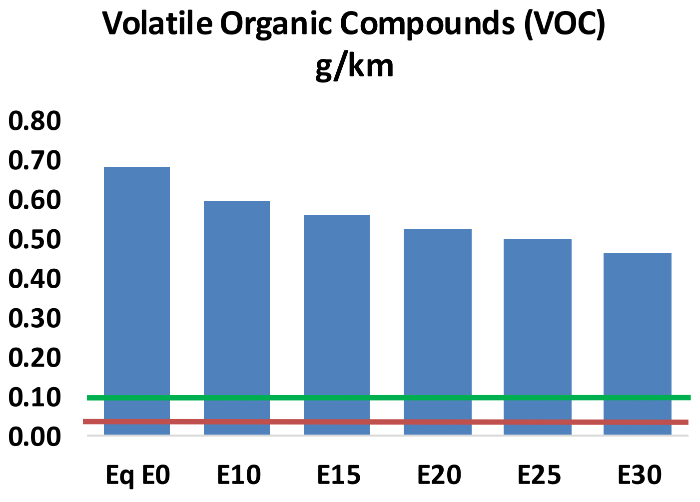
TIER III BIN 125 United States
Euro 6 Standard

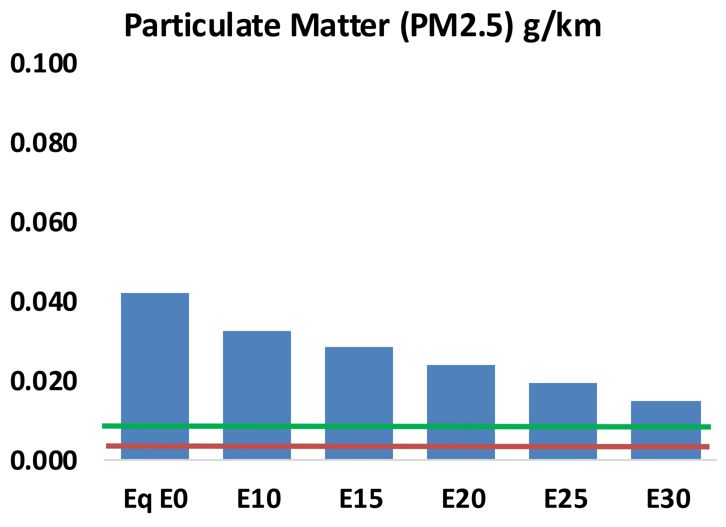












Source: Faro90

Netherlands – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	704,445	654,707	604,969	565,072	525,320	494,872	460,719	-14%	-25%
CO2	28,033,974	27,313,101	26,632,276	26,096,827	25,831,865	25,714,823	25,240,825	-5%	-8%
VOC	103,360	96,697	90,034	84,824	79,696	75,788	70,450	-13%	-23%
NOx	52,857	39,643	37,000	34,873	32,888	30,700	28,384	-30%	-38%
SOx	333	308	283	261	241	219	195	-15%	-28%
NH3	10,778	10,671	10,564	10,615	10,734	10,818	10,888	-2%	0%
N2O	473	470	468	470	480	485	491	-1%	2%
CH4	23,057	23,057	23,057	23,403	23,910	24,348	24,763	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	737	710	682	666	652	642	626	-7%	-12%
Acetaldehyde	2,047	2,349	3,447	5,250	7,152	8,173	9,657	68%	249%
Formaldehyde	7,461	7,253	8,455	9,928	10,401	11,507	12,526	13%	39%
Benzene	1,258	1,209	1,146	1,128	1,118	1,070	1,035	-9%	-11%
PM2.5	1,819	1,615	1,411	1,225	1,041	844	640	-22%	-43%
PM10	4,542	4,033	3,524	3,060	2,599	2,109	1,598	-22%	-43%

Source: Faro90

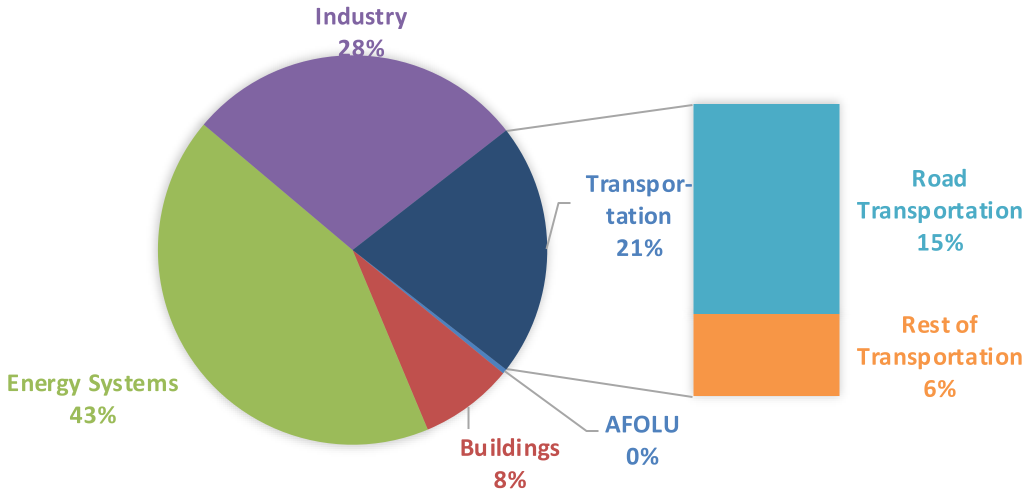
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

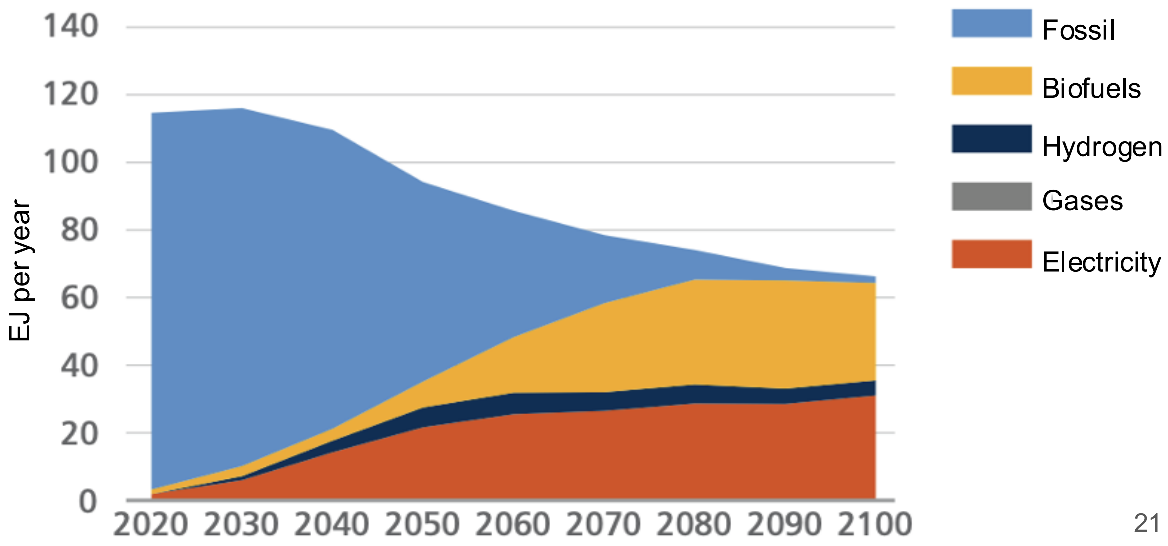
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

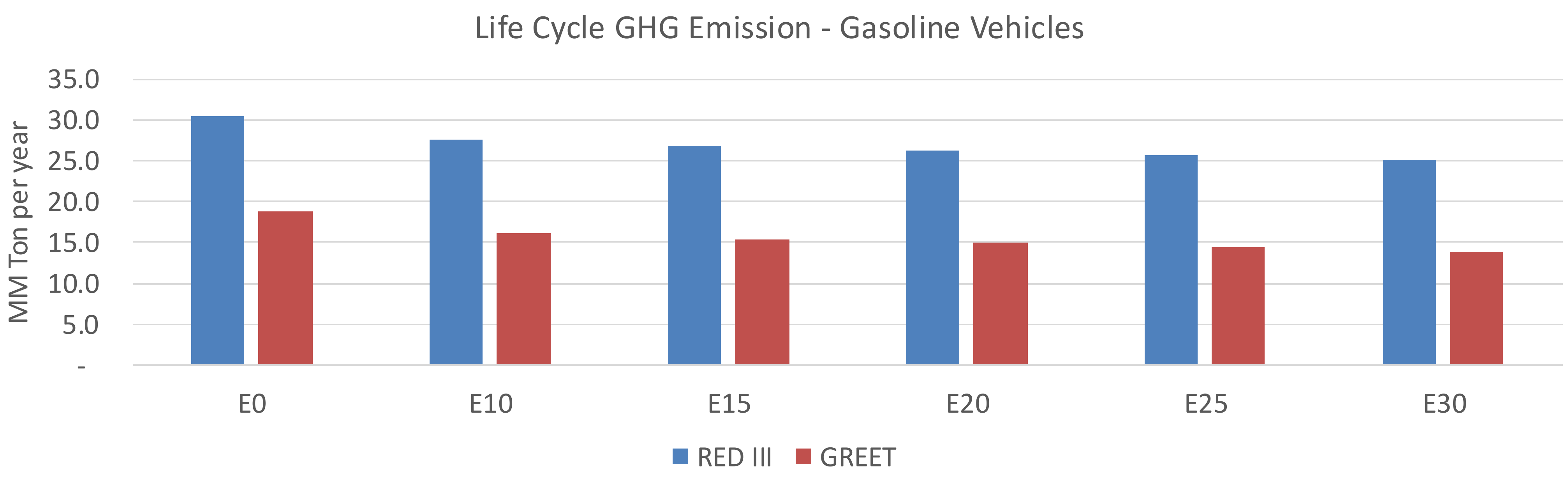
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Netherlands – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	171.1
Target @ 2035 (MMT/year)	96.3
Delta	74.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.0%	18.6%	20.2%	22.9%	26.4%
RED II/III	0.0%	9.2%	11.5%	13.4%	15.3%	17.6%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.5%	4.7%	5.1%	5.8%	6.6%
RED II/III	0.0%	3.7%	4.7%	5.5%	6.2%	7.1%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90